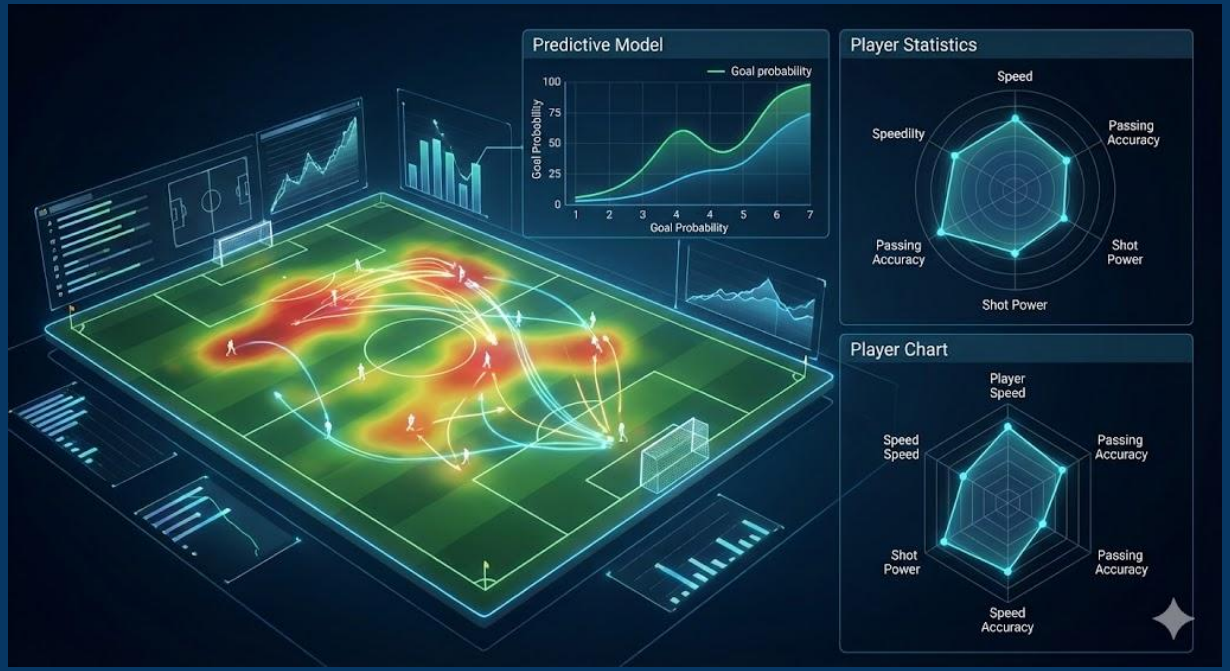
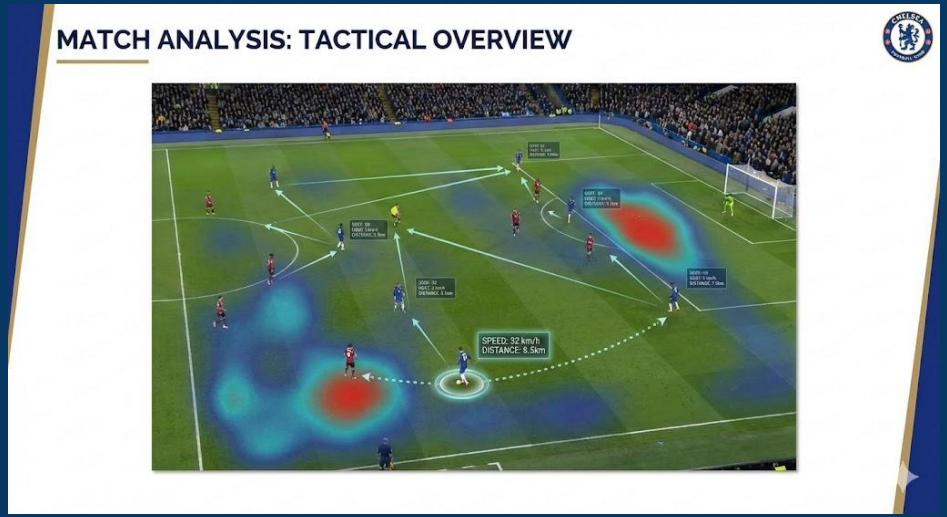
Scouting the Invisible

Could A.I. Find the Next Messi Better Than a Human Scout and Without Bias?



Football/Soccer recruitment used to be about ‘gut feeling’. Now, it’s about data models and algorithms. But are we just automating our own human biases?



The Human Limit

- There are roughly **130,000 professional soccer players** in the world, human **scouts can only watch around 3-5 matches/games per week**.
- **Human scouts are subjective**, they have subconscious **biases** that could be related to a players **nationality, ethnicity, running style, or any other aesthetics**.
- Scouts in most cases **tend to only scout players in higher profile leagues or regions, this causes them to ignore other markets and miss talent in these other regions and lower level leagues.**

Goal: **To Democratize Football/Soccer Recruitment**

- Current **scouting favors rich clubs, famous leagues, and prestigious academies**.
- This project **aims to use data and machine learning to shine a spotlight on "invisible" players in smaller leagues and regions who deserve equal opportunity**.
- My goal is to build an automated "Global Scout" that uses data–not reputation–**to identify undervalued talent**.

Real-World Application (Solution in Action)

Economic Efficiency:

- Brentford FC understand they cannot afford superstars. Instead **they use algorithms to find and buy undervalued players for cheap, develop them, and sell them for huge profit** (ex., buying Neal Maupay for $1.6m, selling for $20m).
- Brighton & Hove Albion **use proprietary data to find players in markets others ignore** (like Moises Caicedo from Ecuador who they bought for $4.5m and sold for $115m), **achieving a massive return on investment.**

Social Impact:

- Research from TMU shows that **when A.I. removes a players name and race from the evaluation, scouts judge purely on skill, reducing racial or pedigree bias**.
- This technology **gives players from poorer nations a fair shot at being discovered** based purely on merit.

Research Question:

- Can an **unsupervised AI algorithm find ‘undervalued assets’**; players who are statistically identical to superstars but are ignored by human scouts due to market bias?

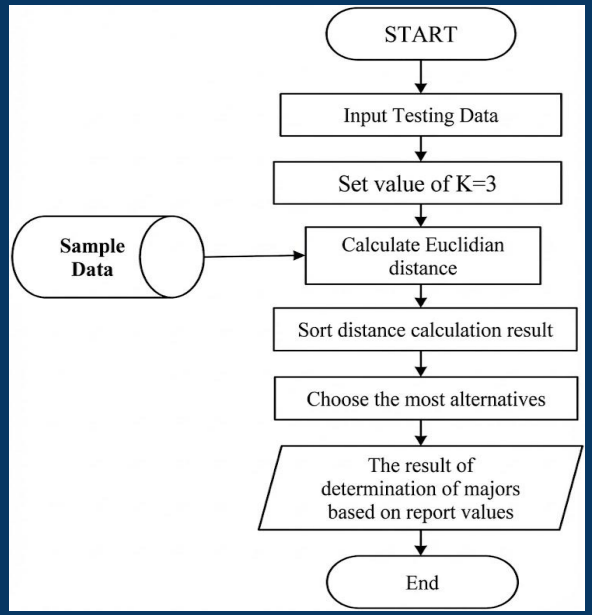
Hypothesis:

I predict that by applying K-Nearest Neighbors (KNN) clustering to a dataset of around 3,000 players, I will identify a target player who:

1. Has a >70% Similarity Score to Erling Haaland (based on 14 performance metrics).
2. Has a Market Value of <$30 Million (approximately 20% of Haaland’s market value).
3. Plays in a league or for a team typically overlooked by major scouts.

Technology & Concept

- This project uses a **Machine Learning algorithm** called **K-Nearest Neighbors (KNN)**.
- Instead of telling the computer which player is a match, I feed it raw data and ask it to discover target players.
- It works exactly **like the Spotify song recommendation algorithm**, that suggests songs that are mathematically similar to what you like. My tool does the same but for Football/Soccer players.



- To calculate similarity, the code uses the **Euclidean Distance** formula.
- It measures the straight-line **distance between two players in a multi-dimensional space** (comparing 14 different variables simultaneously).

$$\text{Euclidean distance} = \sqrt{\sum_{i=1}^{n} \left(x_i - y_i\right)^2}$$

The Tech Stack

Technology used:

- Language: Python 3.0
- Environment: Google Colab Pro (Cloud-based execution)
- Libraries used:

  1.) Pandas - for data manipulation and cleaning.

  2.) Scikit-Learn - For the Nearest Neighbors algorithm and data normalization.

- Data Source: FBref 2024-2025 Season Dataset (via Kaggle)

Performance Metrics:

- Dataset Scope: **The algorithm scans ~3,000 players across 5 top European Leagues**.
- Feature Depth: It **analyzes 14 distinct performance metrics per player** (including xG, Progressive Carries, and Shot Creating Actions).
- Efficiency: The query processing time is < 0.5 seconds per search.
- Similarity Threshold: The model normalizes all the data to a 0-1 scale to ensure fair comparison between different stats.

FBREF

Findings pt. 1

Initial Code to check column titles, load databases and packages

[9]
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```python
# Check available columns
print(list(df.columns))
```

```
['Rk', 'Player', 'Nation', 'Pos', 'Squad', 'Comp', 'Age', 'Born', 'MP', 'Starts', 'Min', '90s', 'Gls', 'Ast',
```

[1]
✓ 1s

```python
# --- CELL 1: SETUP ---
import pandas as pd
from sklearn.neighbors import NearestNeighbors
from sklearn.preprocessing import StandardScaler

# Load the database
# MAKE SURE the file name matches exactly what you uploaded on the left sidebar!
filename = 'players_data-2024_2025.csv'

try:
    df = pd.read_csv(filename)
    print("✅ Success! Database loaded.")
    print(f"   Found {len(df)} players.")
except FileNotFoundError:
    print("❌ Error: File not found.")
    print("   1. Click the Folder icon 📁 on the left.")
    print("   2. Drag and drop your .csv file there.")
    print("   3. Make sure the filename matches the code exactly.")
```

```
✅ Success! Database loaded.
   Found 2854 players.
```

Findings pt. 2

Code used to train the model, establish the metrics. This is also cleaning, filtering, and normalizing the dataset.

[10]

```python
# --- CELL 2: TRAIN THE AI (UPDATED) ---

# EXPANDED STATS LIST
# We are now using 14 distinct metrics for a full "360 Degree" view of the player
features = [
    'Gls', 'Ast',       # Scoring
    'xG', 'npxG',       # Expected Goals (Quality of chances)
    'PrgC', 'PrgP',     # Progression (Carries & Passes)
    'Tkl', 'Int',       # Defense (Tackles & Interceptions)
    'Blocks', 'Clr',    # Defense (Blocks & Clearances)
    'AerWon',           # Physical (Aerial Duels Won)
    'SCA',              # Shot Creating Actions (Creativity)
    'Cmp%',             # Passing Accuracy
    'DrbSucc%',         # Dribbling Success %
    'Touches',          # Involvement in the game
    'Min'               # Playing Time
]

# Clean the data
# We check if these columns exist first to avoid crashing
available_features = [col for col in features if col in df.columns]

# Filter data
df_clean = df[df['Min'] > 500].copy()
df_clean[available_features] = df_clean[available_features].fillna(0)

# Normalize
scaler = StandardScaler()
scaled_features = scaler.fit_transform(df_clean[available_features])

# Train Model
model = NearestNeighbors(n_neighbors=6, metric='euclidean')
model.fit(scaled_features)

print(f"✅ AI Model upgraded! Now tracking {len(available_features)} different stats.")
```

```
✅ AI Model upgraded! Now tracking 14 different stats.
```

Findings pt. 3

Code used to create the scouting tool which can find target players based on similarity score

[12]
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```python
# --- CELL 3: SCOUTING TOOL (CLEAN LIST VERSION) ---
def find_bargain_players(player_name):
    # 1. Find the target player in the database
    target = df_clean[df_clean['Player'].str.contains(player_name, case=False, na=False)]

    if target.empty:
        print(f"❌ Player '{player_name}' not found.")
        return

    # Get the target's stats for calculation
    target_idx = target.index[0]
    target_stats = scaled_features[df_clean.index.get_loc(target_idx)].reshape(1, -1)

    # Get the target's info for display
    t_player = target.iloc[0]

    print(f"\n🔎 ANALYZING TARGET: {t_player['Player']}")
    print(f"   Team: {t_player['Squad']}")
    print(f"   Key Metric: {t_player['xG']} xG (Expected Goals)")
    print("=" * 60)

    # 2. Ask AI for the 5 closest matches
    distances, indices = model.kneighbors(target_stats)

    print(f"🤖 AI RECOMMENDATIONS (Sorted by Similarity):")

    for i in range(1, len(indices[0])):
        idx = indices[0][i]
        sim_player = df_clean.iloc[idx]

        # Calculate similarity %
        similarity = max(0, 100 - (distances[0][i] * 15))

        print(f"#{i}: {sim_player['Player']}")
        print(f"    Team: {sim_player['Squad']} ({sim_player['Comp']})")
        print(f"    Similarity Score: {similarity:.1f}%")

        # Display key stats (excluding AerWon as requested)
```

Findings pt. 4

Output/Results, giving a list of 5 target players who have a close high similarity score to Erling Haaland in this case.

Hypothesis Confirmed: Undervalued Talent Identified

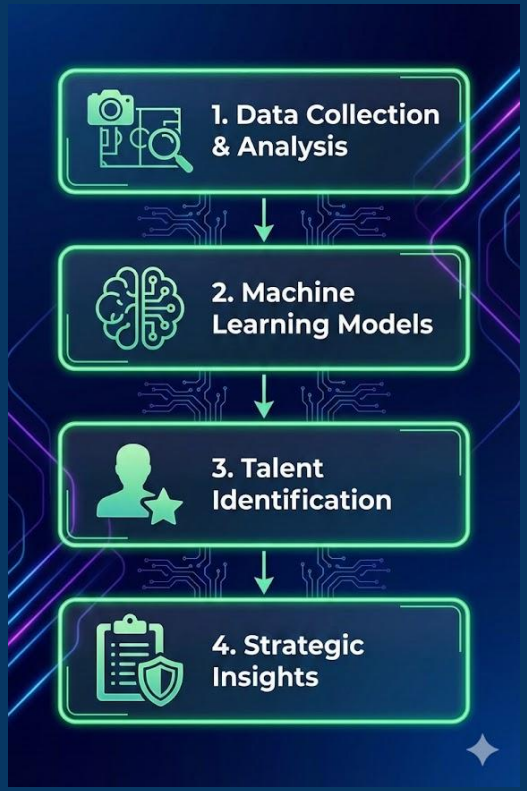
- The algorithm identified Serhou Guirassy from Dortmund and Moise Kean from Fiorentina as the closest statistical twins to Erling Haaland of Man. City.
- Despite playing in a lower level league and a lower level team both of these players matched Haalands output in key metrics like xG, Shot Creating Actions, and Progressive Carries as an example.
- But the most important number isn't on the code output; it's the price tag. Haaland’s worth 180 million. Serhou Guirassy costs just 40 million. My algorithm just found nearly an 80% like for like replacement for nearly 20 % of the cost.

```
# --- TYPE THE PLAYER NAME BELOW ---
find_bargain_players("Haaland")
```

```
🔎 ANALYZING TARGET: Erling Haaland
   Team: Manchester City
   Key Metric: 22.0 xG (Expected Goals)
============================================================
🤖 AI RECOMMENDATIONS (Sorted by Similarity):
#1: Serhou Guirassy
    Team: Dortmund (de Bundesliga)
    Similarity Score: 78.9%
    Stats: 22.7 xG | 53 Shot Actions | 25 Carries
----------------------------------------
#2: Moise Kean
    Team: Fiorentina (it Serie A)
    Similarity Score: 71.2%
    Stats: 19.4 xG | 57 Shot Actions | 42 Carries
----------------------------------------
#3: Yoane Wissa
    Team: Brentford (eng Premier League)
    Similarity Score: 58.3%
    Stats: 18.5 xG | 69 Shot Actions | 60 Carries
----------------------------------------
#4: Alexander Sørloth
    Team: Atlético Madrid (es La Liga)
    Similarity Score: 55.4%
    Stats: 16.8 xG | 30 Shot Actions | 32 Carries
----------------------------------------
#5: Ante Budimir
    Team: Osasuna (es La Liga)
    Similarity Score: 52.8%
    Stats: 18.3 xG | 46 Shot Actions | 23 Carries
----------------------------------------
```

In-Scope

- Statistical Clustering: I focused this project strictly on Quantitative Analysis using unsupervised machine learning (KNN).
- Performance Metrics: The model evaluates players based on 14 objective data points from the 2024/25 season.
- Economic Identification: This project is limited to identifying statistical similarity to high potential "undervalued assets" in the transfer market.



Out-of-Scope

- Qualitative Analysis (Eye-Test): This project does not use Computer Vision or video analysis.
- Psychological Factors: the dataset does not measure 'soft skills' like leadership, adaptability, or injury proneness.
- Live Market Dynamics: the dataset does not measure exactly player value, they are estimated based on static data not live market price.

Conclusion: This tool is designed to filter the list of 3,000 players down to a shortlist of 5 based on a similarity score which is based on 14 performance metrics. It is a Decision Support System, not a replacement for human judgment.

Sources

Academic Research:

- Dubois, L.-E., & Walzak, L. (2025). Blind scouting: Using artificial intelligence to alleviate bias in selection. *Personnel Review*.

Technical Resources:

- Sidorowicz, H. (2025). *Football Players Stats (2024-2025)* [Data set]. Kaggle. https://www.kaggle.com/datasets/hubertsidorowicz/football-players-stats-2024-2025
- *Scikit-learn: Machine Learning in Python*. (2024). Scikit-learn developers.